

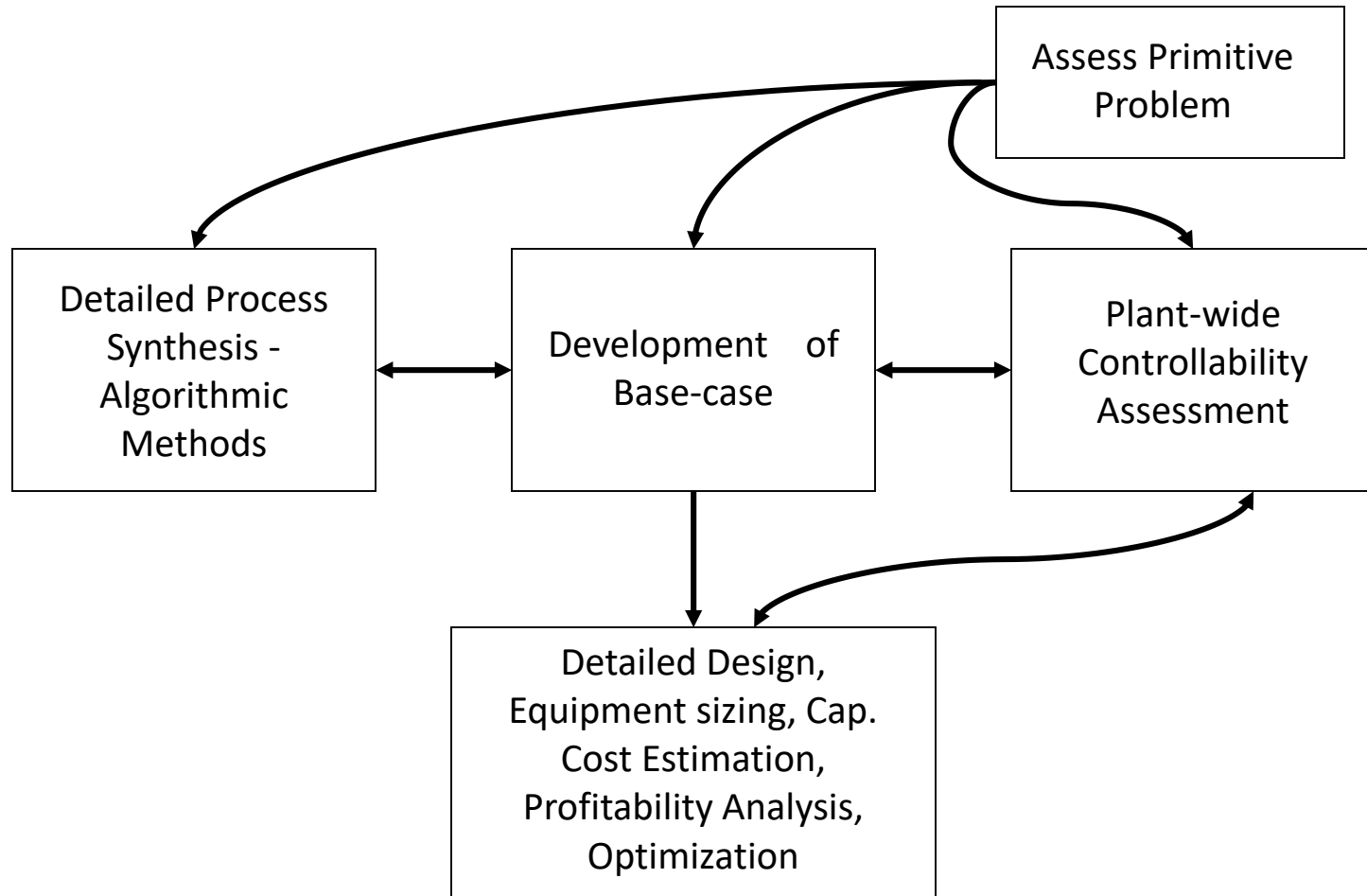


Lecture-2

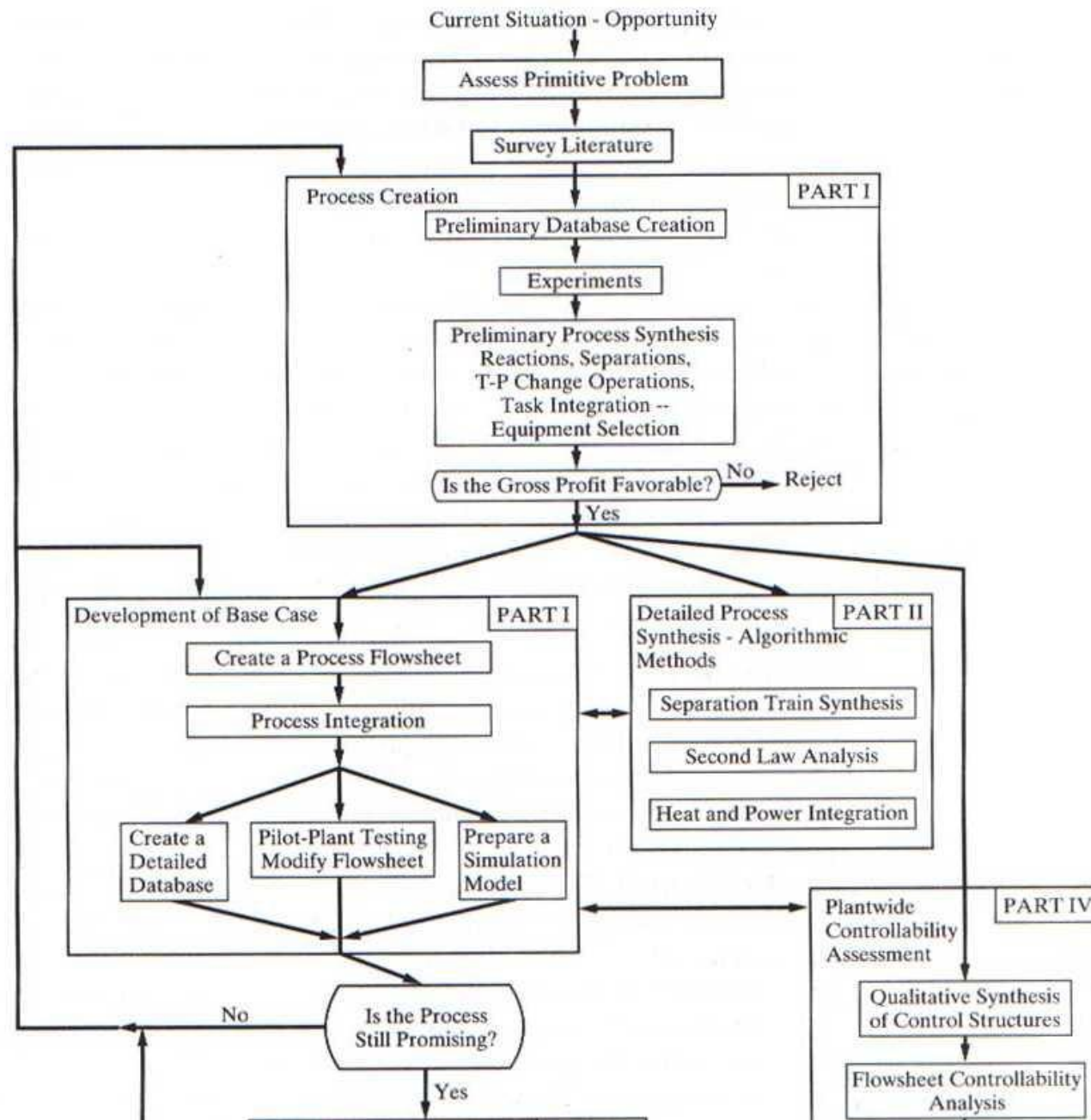
Hierarchical Approach to Conceptual Design

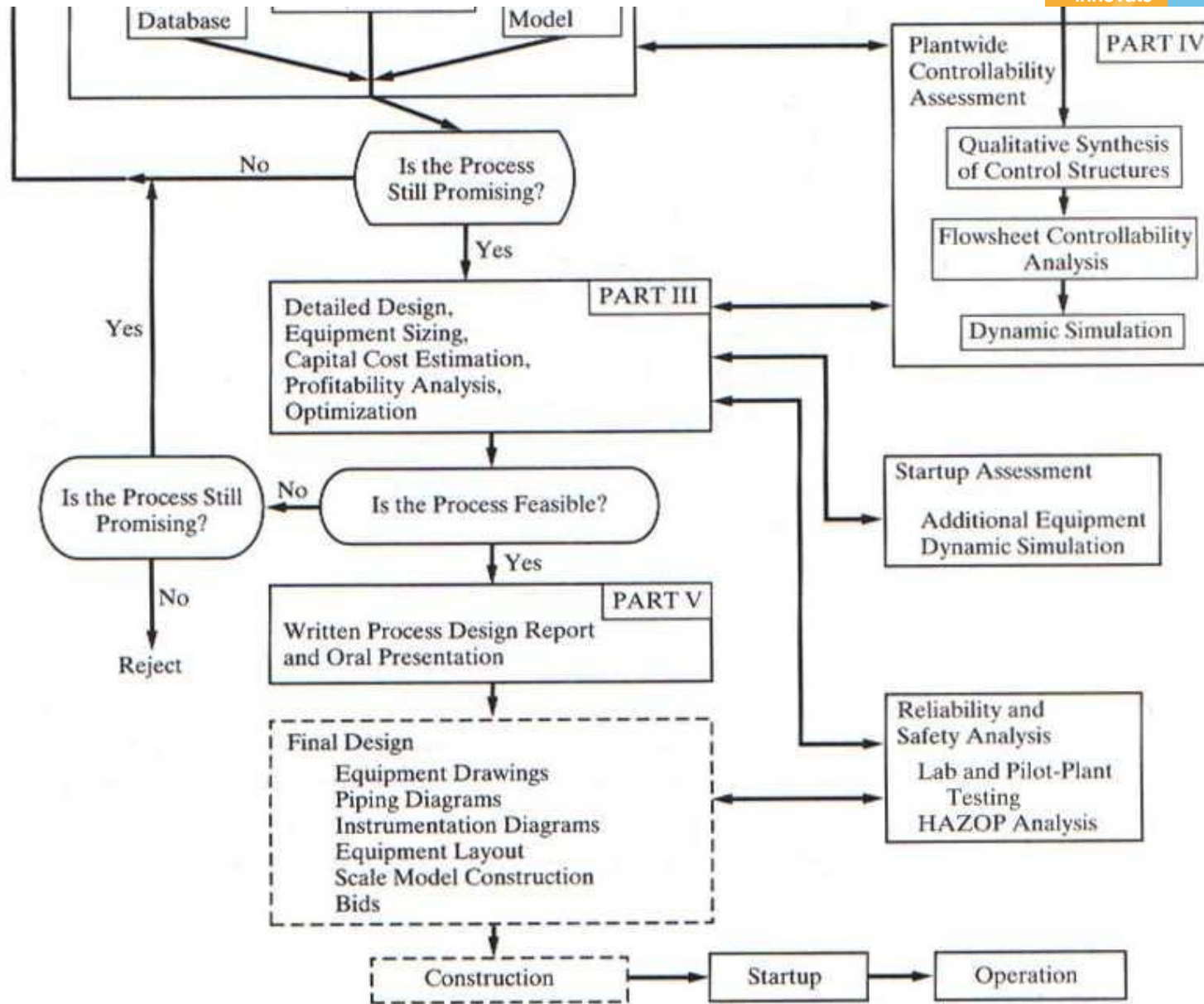
06-08-2026

Steps in Process Design and Retrofit



Source Ref: Seider et al., 2010







Preliminary Process Synthesis

- Synthesis of chemical processes involves:
 - Selection of processing mode: continuous or batch
 - Fixing the chemical state of raw materials, products, and by-products, noting the differences between them.
 - Process operations (unit operations) - flowsheet building blocks
- Synthesis steps -
 - Eliminate differences in molecular types
 - Distribute chemicals by matching sources and sinks
 - Eliminate differences in composition
 - Eliminate differences in temperature, pressure and phase
 - Integrate tasks (combine tasks into unit operations)

Synthesis Step



Synthesis Step

- ① Eliminate differences in molecular types
- ② Distribute chemicals by matching *sources* and *sinks*
- ③ Eliminate differences in composition
- ④ Eliminate differences in temperature, pressure and phase
- ⑤ Integrate tasks (combine *tasks* into *unit operations*)

Process Operation

Chemical reaction

Mixing

Separation

Temperature, pressure and phase change

Source Ref: Seider et al., 2010



Outline

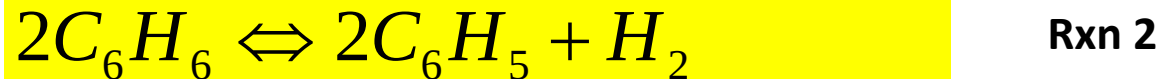
- Hierarchical Approach to Conceptual Design: HDA Case Study
 - Simplified flowsheet for the separation process
 - Recycle structure of flowsheet
 - Input-Output Structure of Flowsheet
 - Hierarchy of Decisions



Hierarchical Approach to Conceptual Design

Example: Hydrodealkylation of toluene (HDA Process)

- To produce benzene



Reaction temperatures for homogeneous reactions: 1150 – 1300°F

- If $T < 1150^\circ\text{F}$ the reaction rate is very slow
- If $T > 1300^\circ\text{F}$ a significant amount of hydrocracking takes place
- Pressure 500 psia (≈ 34 atm)
- Excess hydrogen (H_2 : aromatics = 5:1)
- Reactor effluent gas must be rapidly quenched to 1150°F



Boiling points

Boiling Point (°F)	
Diphenyl	491
Toluene	232
Benzene	176.2
Methane	-258.68
Hydrogen	-423.182







Energy Integration

- Is the process flow sheet very realistic?
- In the last decade (1978), a new design procedure has been developed
 - that makes possible to find the minimum heating and cooling loads for a process
 - and the Heat Exchanger Network Synthesis (HENS) that gives the 'Best' energy integration

Energy Integrated Flow sheet

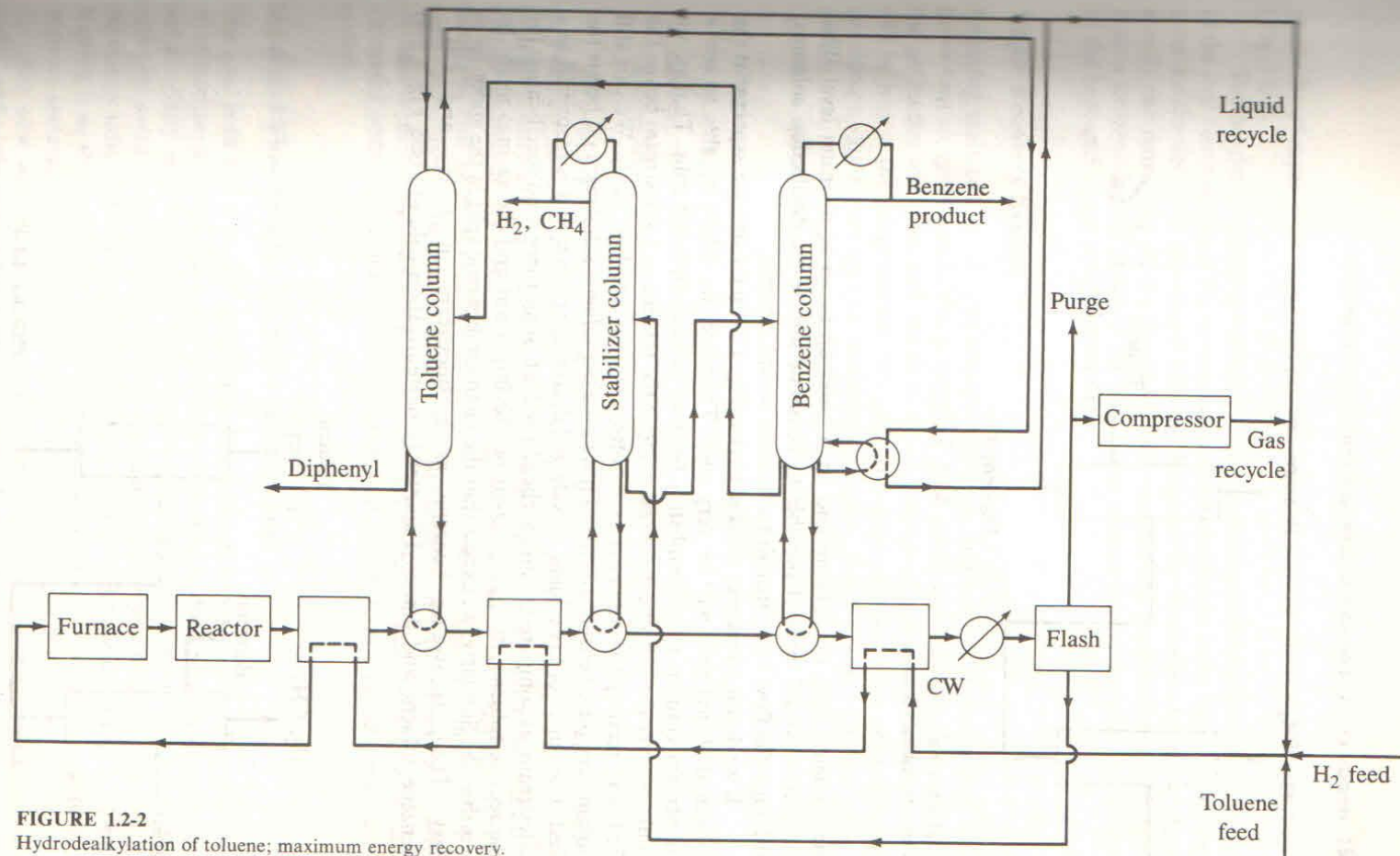


FIGURE 1.2-2

Hydrodealkylation of toluene; maximum energy recovery.

Fig. 2



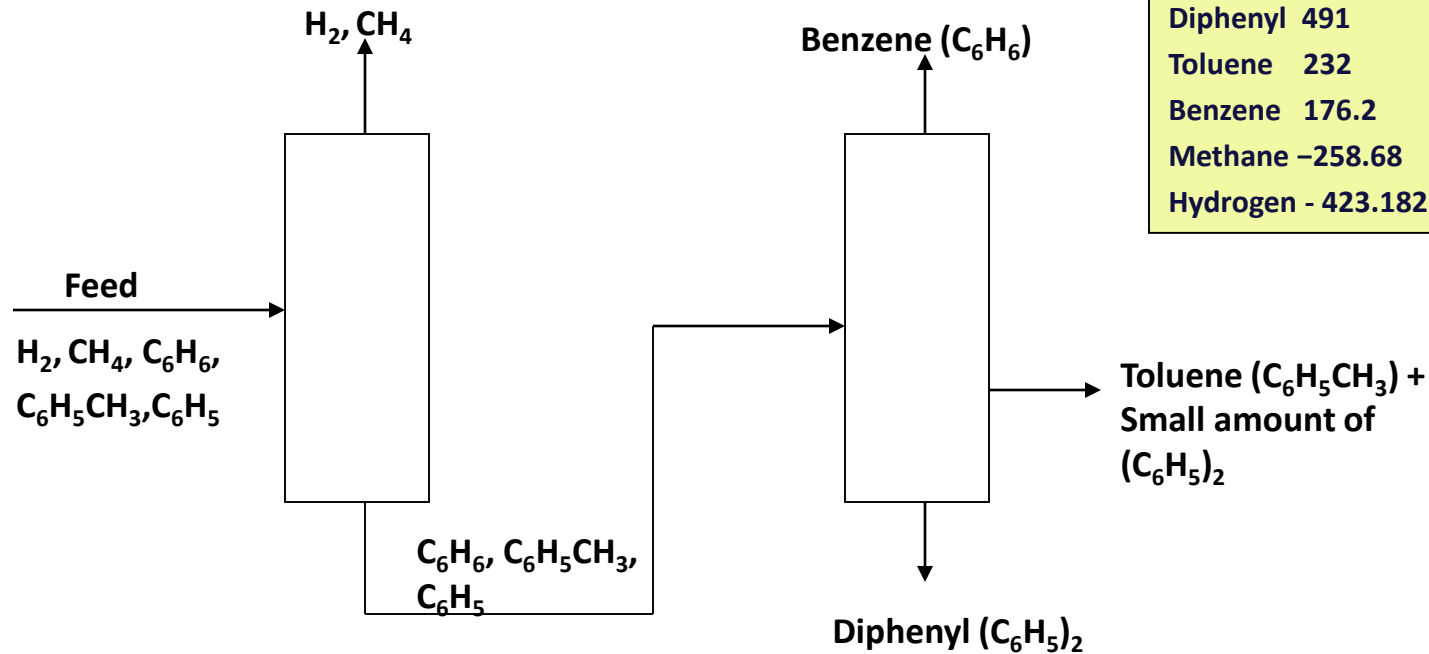
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- Energy Integration flow sheet is more complicated
 - many more interconnection
- Moreover to apply the Energy Integration (HENS) analysis
 - we must know the flow rate and composition of every process stream i.e. all the process heat loads including those of the separation system as well as all the stream temperatures
- Since we need to fix almost all the flow sheet before we can design the Energy Integration system
 - since it adds the greatest complication to the process flow sheet
 - we consider the Energy Integration Analysis (HENS) as last step in our process design procedure



Distillation Train

- We could recover the benzene as overhead
- Remove toluene as the side-stream (below the feed), and recover the diphenyl as a bottom stream

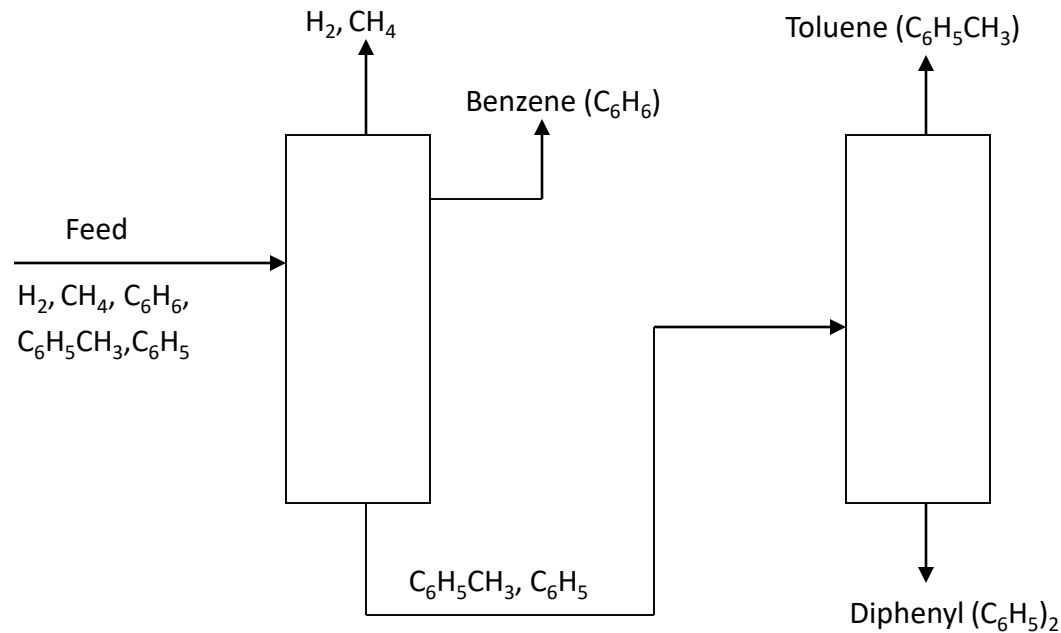


Boiling Point (°F)	
Diphenyl	491
Toluene	232
Benzene	176.2
Methane	-258.68
Hydrogen	-423.182

Fig. 3



Contd..



Boiling Point (°F)	
Diphenyl	491
Toluene	232
Benzene	176.2
Methane	-258.68
Hydrogen	-423.182

Fig. 4



Contd..

- It might be cheaper than using the configuration shown in the original flow sheet (Fig. 1)
- The heuristics (design guidelines) for separation systems require
 - A knowledge of the feed composition of the stream entering the distillation train
- Thus before we consider the decisions associated with the distillation train, we must specify the remainder of the flow sheet and estimate the process flows
- For this reason we consider the design of the distillation train before we consider the design of the heat-exchanger network



Vapor Recovery System (VRS)

- Complete separation (of aromatics and light gases) in a flash drum NOT POSSIBLE!
 - therefore that some of the aromatics will leave with the flash vapor (H_2 and CH_4 lighter gases)
- Moreover some of those aromatics will be lost in the purge stream
- It is possible to recover those aromatics by installing a VRS either on the flash vapor stream or on the purge stream



Contd..

- As a VRS, one of the following can be used
 - Condensation (high pressure or low temperature or both)
 - Absorption
 - Adsorption
 - A membrane process
- To find out the economic feasibility of the VRS
 - we must estimate the flow rates of aromatics lost in the purge as well as the H_2 and CH_4 flow in the purge
- Hence before we consider the necessity and / or the design of a VRS
 - we must specify the remainder of the flow sheet and we must estimate the process flows



Contd..

When do we consider designing of VRS?

We consider the design of the VRS before that for the liquid separation system



Simplified Flowsheet for the Separation Systems

- Our goal is to find a way of simplifying flowheets
- It is obvious that Fig.1 is much simpler than the figure in which energy integration (HENS) is included
 - because of which it was decided that the EIA be carried out at the end (after distillation train is finalized)
- Similarly, since we have to know that the process flow rates to design the VRS and LRS
 - it was decided to consider these design problems just before the energy integration

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- The connections between the VRS and LRS shown in Fig. 5

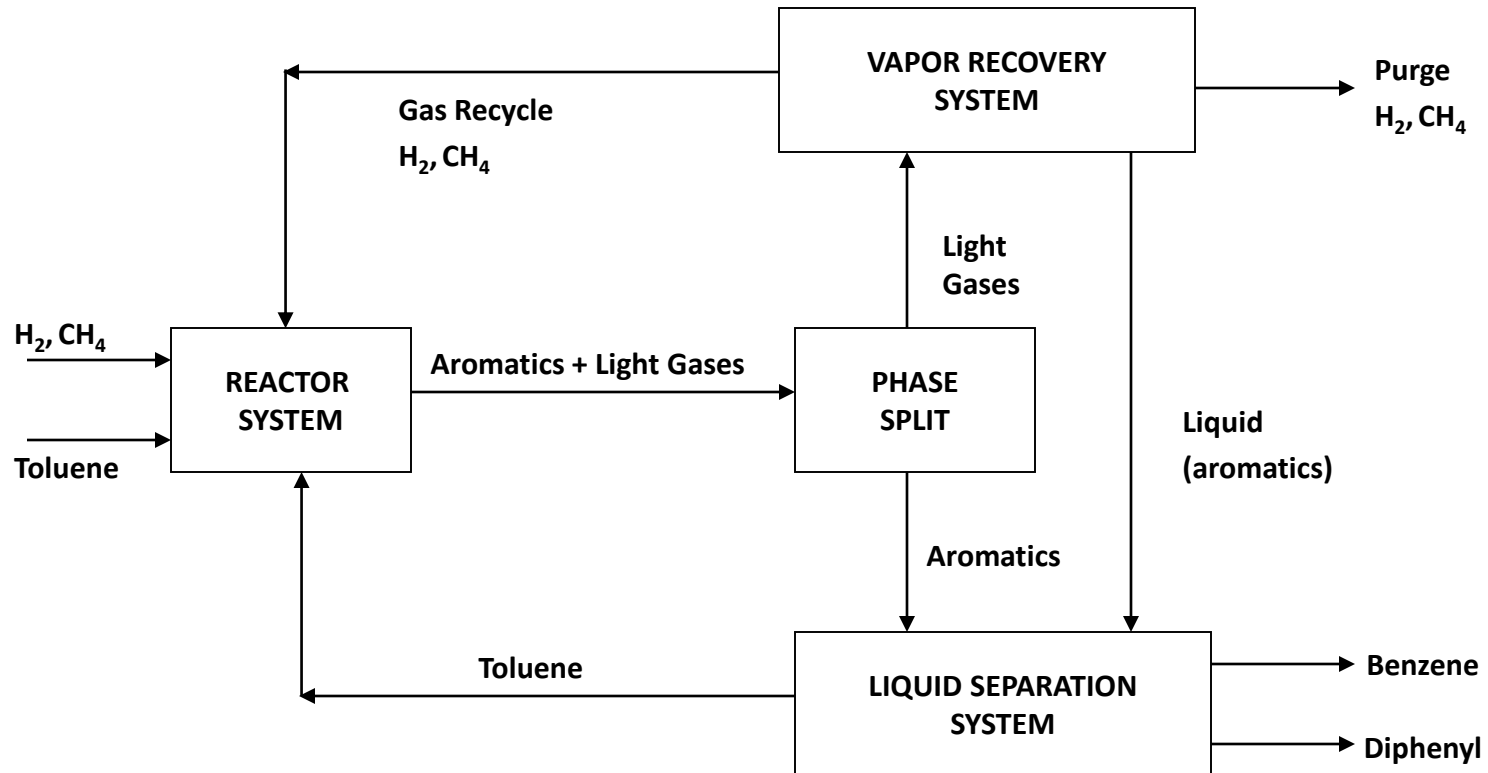


Fig. 5

Recycle structure of the flow sheet

A simplified flow sheet for the process is shown in Fig. 6

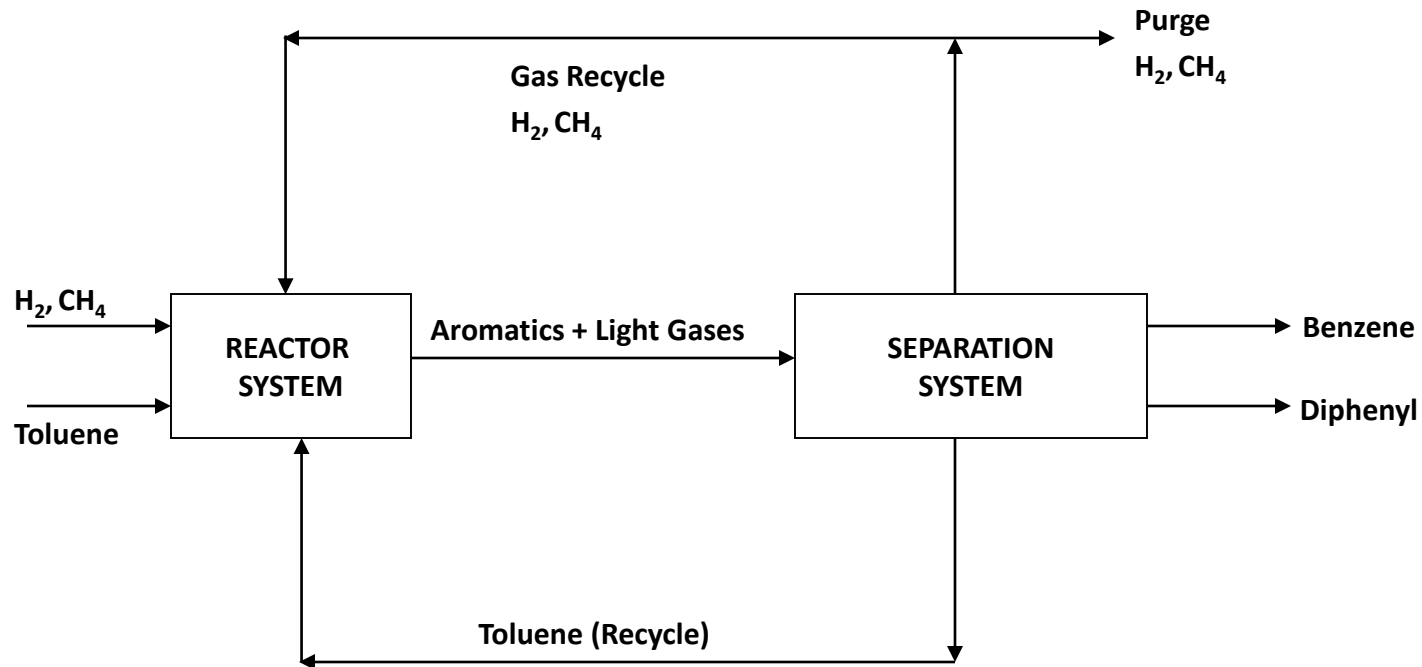


Fig. 6



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- Use this simple representation
 - to estimate the recycle flows
 - their effect on the reactor cost, and
 - the cost of gas recycle compressor, if any
- For example, we can study:
 1. The factors that determine the no. of recycle streams
 2. Heat effects in the reactor
 3. Equilibrium limitations in the reactor, etc.

Can we still think of simplifying the flowsheet?



Input-Output structure of the flowsheet

- Since raw material costs normally fall in the range from 33-85% of the total product costs
 - the overall material balance are the dominant factors in the design

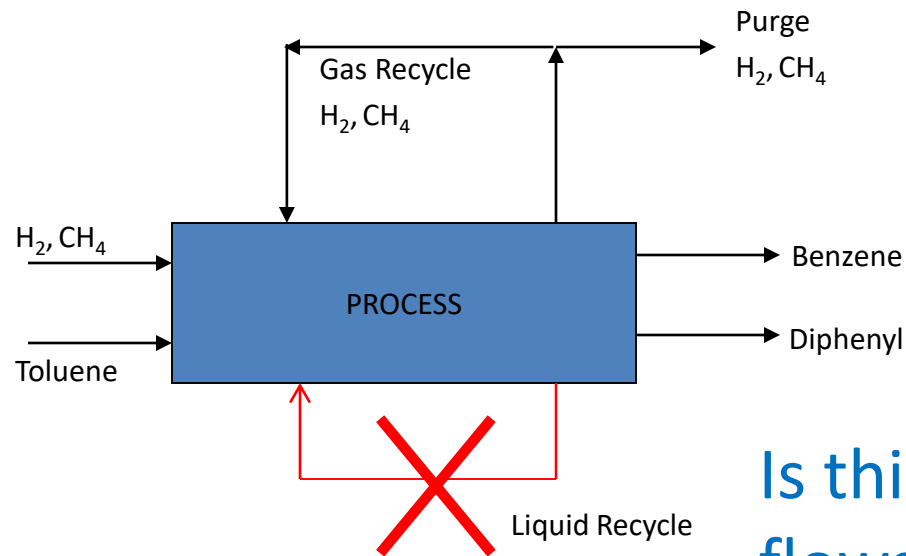


Fig. 7



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- Also we do not want to spend any time investigating the design variables in the ranges
 - where the products and by products are worth less than the raw materials
- Thus, we consider the Input-output structure of the flow sheet and the decisions that affect this structure before we consider any recycle streams
- By successively simplifying a flowsheet, we can develop a general procedure for attacking design problems



Hierarchy of Decisions

- A systematic approach to process design by reducing the design problem to a hierarchy of decisions:
 1. Batch vs Continuous
 2. Input-Output structure of the flow sheet
 3. Recycle structure of the flow sheet
 4. General structure of the separation system
 - a) Vapor liquid system
 - b) Liquid separation system
 5. Energy Integration Analysis (HENS)



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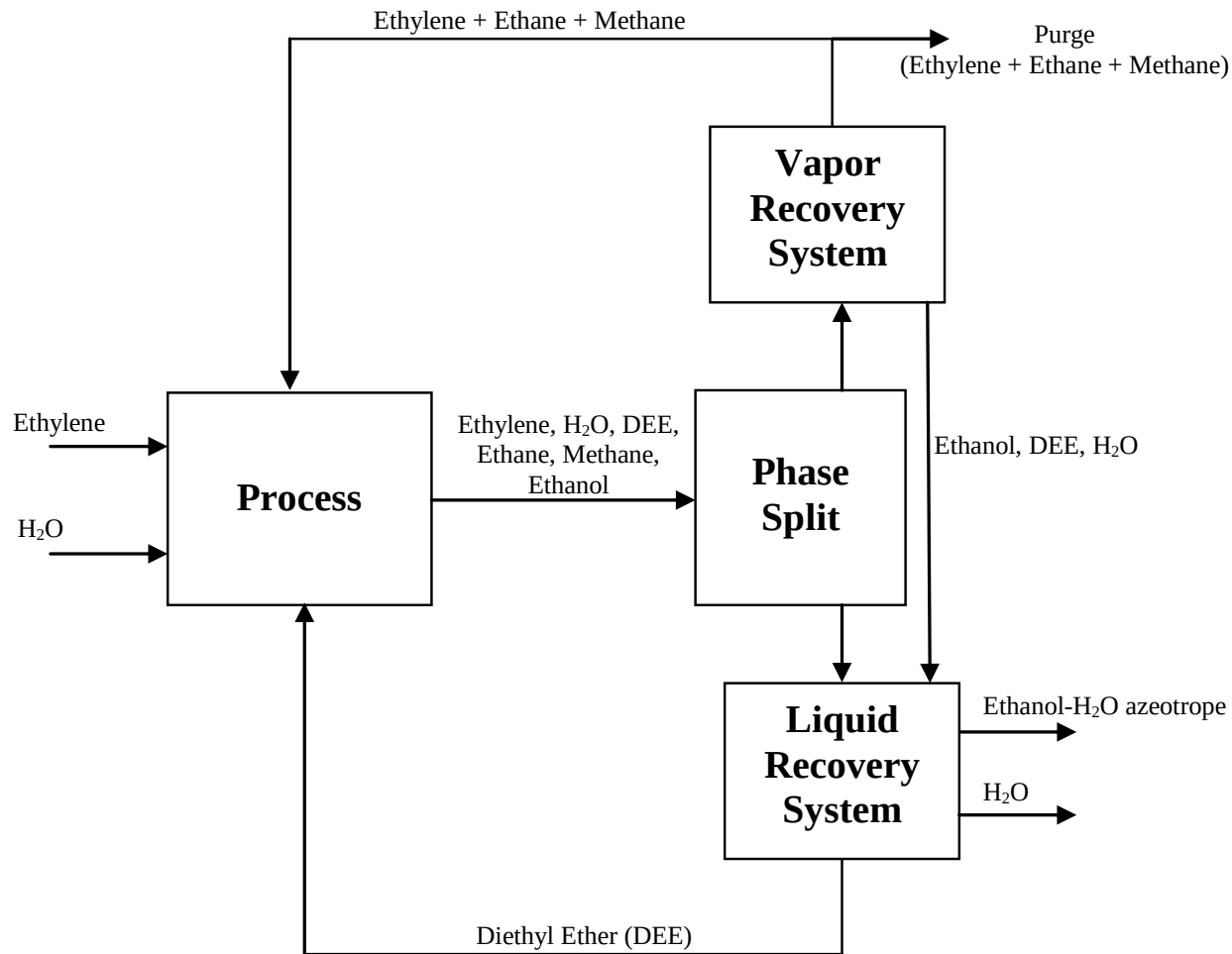
- One great advantage of this approach to design is:
 - It allows us to calculate equipment cost
 - to estimate costs
- Then if the potential profit becomes negative at some level
 - look for a process alternative or ,
 - terminate the design project without having to obtain a complete solution to the problem
- Another advantage of this procedure:
 - As we make about the structure of the flow sheet at various levels
 - We know that if we change these decisions, we will generate process alternatives
- The goal of a conceptual design is to **find the best alternative**



Example

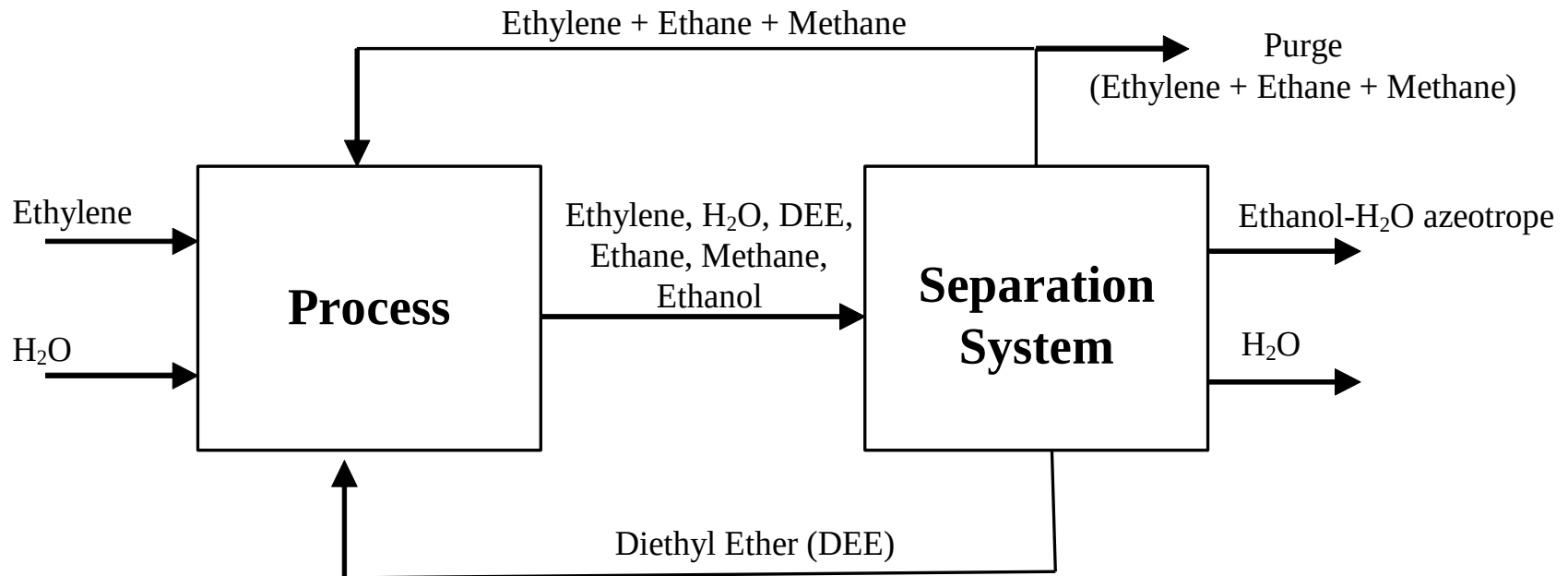
- Ethanol is produced by the hydration of ethylene. The primary reactions for ethanol synthesis are given below:
$$\text{Ethylene} + \text{H}_2\text{O} \rightleftharpoons \text{Ethanol}$$
$$2 \text{ Ethanol} \rightleftharpoons \text{Diethyl ether} + \text{H}_2\text{O}$$
- Initially, the feed (90% ethylene, 8% ethane, and 2% methane) and water are heated by passing through the primary heater. This heated feed is sent to the reactor. The reaction takes place at 560 K and 69 bar. The fractional conversion of ethylene in the reactor is 0.07. The reactants and products are sent to the separator where gaseous and liquid products and reactants are separated. All gaseous products and reactants are scrubbed in a scrubber. Unconverted ethylene and inert gases (ethane and methane) are recycled back. To avoid the accumulation of inert components, some amount of recycled stream is purged. The liquid products and the bottom products of scrubber are sent to the series of distillation columns where side product diethyl ether and water are separated out. The diethyl ether is recycled back and mixed with the feed stream. Ethanol-water azeotrope is produced from the final distillation column. Draw the following:
 - General structure of the flow sheet
 - Recycle structure of the flow sheet
 - Input-output structure of the flow sheet

General structure



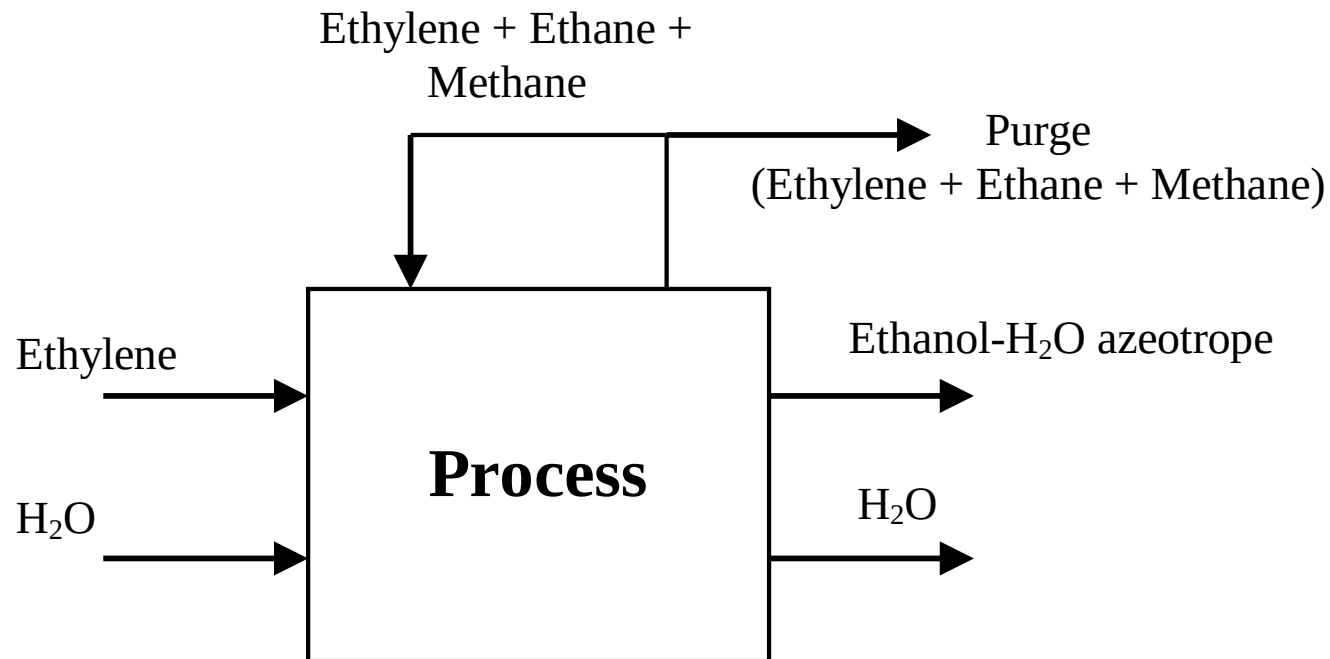


Recycle structure





Input-output structure





Problem

- In a formox process, a mixture of metal oxides, consisting of Fe, MO, and V, is used as a catalyst for the partial oxidation of methanol to formaldehyde. Different reactions taking place in the reactor are as follows:

$$\text{CH}_3\text{OH} + 1/2 \text{O}_2 \rightarrow \text{HCHO} + \text{H}_2\text{O}$$

$$\text{CH}_3\text{OH} + 3/2 \text{O}_2 \rightarrow \text{CO}_2 + \text{H}_2\text{O}$$

$$\text{CH}_3\text{OH} \rightarrow \text{CO} + 2\text{H}_2$$

$$\text{CH}_3\text{OH} \rightarrow \text{CH}_4 + 1/2 \text{O}_2$$

$$2\text{CH}_3\text{OH} \rightarrow (\text{CH}_3)_2\text{O} + \text{H}_2\text{O}$$
- Methanol feed is passed through a steam-heated evaporator. Ambient air, containing 0.011 kg moisture/kg dry air, is heated by a product gas stream, leaving the reactor, and mixed with methanol vapors in a static mixer.

The gaseous feed, containing 8.4% methanol (by volume on a wet basis), employs excess air and is a flammable mixture. It is passed at 170 kPa through catalyst-filled tubes in a heat-exchanging type reactor. Downtherm – A circulates outside the tubes and removes the heat of reaction. Reactions take place isothermally at 613 K (340 °C). Steam is generated at 6 bar in a waste heat boiler (WHB) by condensing Downtherm – A vapors. Gas mixture from the reactor is cooled to 383 K (110 °C) in a heat exchanger by heating incoming air.

The gas mixture then enters the bottom of the absorber. Formaldehyde concentration at the bottom of the absorber is controlled at 37% (by mass) by freshwater, entering at the top at 303 K (50 °C). Tail gas leaves the absorber at 120 kPa and 323 K (50 °C) at the top. Assume 99% conversion of methanol with 90% yield of formaldehyde. Draw the following hierarchy of flow sheets:

General structure

Recycle structure

Input-Output structure



Problem

- Cumene ($\text{C}_6\text{H}_5\text{C}_3\text{H}_7$) is manufactured by reacting benzene (C_6H_6) with propylene (C_3H_6) in a fixed-bed catalytic reactor. A gas stream containing butane and propylene and a liquid stream containing essentially pure benzene are fed to the reactor. Fresh benzene and recycled benzene (both at 77°F) are mixed and passed through a heat exchanger where they are heated by the reactor effluent before being fed to the reactor. The reactor effluent enters this exchanger at 400°F and leaves at 200°F . The pressure in the reactor is sufficient to maintain the effluent stream as a liquid. After being cooled in the heat exchanger, the reactor effluent is fed to a distillation column. All of the butane and unreacted propylene are removed as overhead products from the column, and the cumene and unreacted benzene are removed as bottom products and fed to a second distillation column where they are separated. The benzene leaving from the top of the second column is recycled and mixed with the fresh benzene feed.

Show the following hierarchy of flow sheets:

- General structure
- Recycle structure
- Input-Output structure



Thank You